Introduction to Probability and Statistics

(EE 354 / CE 361)

Practice Problems

Exercise No 1

Conditional Probability and Bayes Rule

A Habib University student commutes to the university (from her home) using one of the following options:

- **Option 1:** Takes the home car to the university 50% of the time. Under this option, she has an 85% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 15 to 25 minutes.
- **Option 2:** A friend gives her a ride 25% of the time. Under this option, she has a 50% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 10 to 15 minutes.
- **Option 3:** She takes a ride share option 25% of the time. Under this option, she has a 50% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 20 to 25 minutes.

(a) Calculate the probability that the student on a given day reaches the university late. 5 Marks

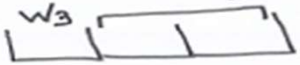
(b) Given that she reached the university in time on a particular day, what is the probability that her friend gave her a ride to the campus that day? 5 Marks

(c) Using the Total Expectation Theorem, calculate the expected commute time it takes for her to reach the campus. 5 Marks

Solution Q#3

(a) Activity A

$^7C_2$ (1)



= 21 (1.5)

without any Restriction

One of these Combination $W_3 W_1 W_2$ = 1 (1.5) NOT allowed

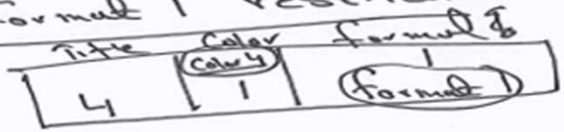
Valid Workshop Selections = 21 − 1 = 20 (1)

(b) Activity B Lets find distinct valid presentation for Title 1 to 4 (Title 1, 2, 3, 4)



= 48 (1 pt)

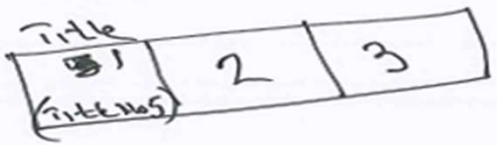
Enforcing format 1 restriction (1 pt)



= 4 ways → Not allowed.

⇒ Valid (distinct) presentations using title 1, 2, 3, 4 = 48 − 4 = 44 (1 pt)

Now, for Title 5



= 6 valid presentations for Title 5.

Total valid Presentations = 44 + 6 (1 pt)

= 50 (1 pt)

Exercise No 2

Distribution & Conditional Distribution with Discrete Random Variables

An HU student was intrigued in creating a 3-sided biased die (with numbers 1, 2, 3) and came up with an interesting idea. He developed a fair 6-sided die (each of the 6 sides with equal probability) and instead of numbers 1 to 6 appearing on each side, he engraved Number 1 on one of the 6 sides, Number 2 on two of the 6 sides and Number 3 on the three remaining sides. He rightly argued that this was a 3-sided biased die.

You roll this 3-sided biased die two times, noting the value of the **DRV(discrete random variable)** X (X is the number of times the Number 3 appears in the two rolls of the biased 3-sided die).

(a) Calculate and plot the P.M.F., $P_X(X)$ for the DRV (discrete random variable) X. 4 Marks
(*Hint – X can take on values* $0, 1,$ *and* 2.)

We now define a new DRV Y, where the conditional PMF $P(Y|X = x)$ is a discrete uniform distribution taking on values x to 3.(Therefore, if R.V $X = 0$, then the conditioned R.V $Y|X = 0$ has a discrete uniform distribution taking on values $0, 1, 2, 3$, each with probability 0.25. Similarly, if the R.V $X = 2$, then the conditioned R.V $Y|X = 2$ has a discrete uniform distribution taking on values $2, 3$.)

(b) Calculate and plot the conditional P.M.F. $P_{Y|X}(y|x)$ for the D.V Y. 8 Marks

(*Hint – There will be three conditional PMFs* $P_{Y|X}(y|x)$ *corresponding to* $X = 0, 1,$ *and* 2.)

(c) Calculate and plot the P.M.F. $P_Y(y)$ for the random variable Y. 8 Marks

(*Hint – Use the above answers and the Total Probability Theorem to calculate* $P_Y(y)$ *for all values of* y, *where* $y = 0, 1, 2, 3$.)

(a) 6-sided die has No 1 on 1 side, No 2 on 2 sides and No 3 on 3 sides

3-sided die

$\Rightarrow$ 1 w.p $1/6$
2 w.p $2/6$
3 w.p $3/6$

2 pt

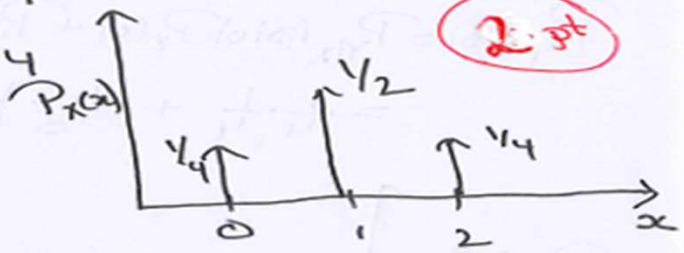
$P(\text{No } 3) = 1/2$, Probability $P(\text{Not getting } 3) = 1/2$.

RV X is No of times 3 appears in 2 Rolls

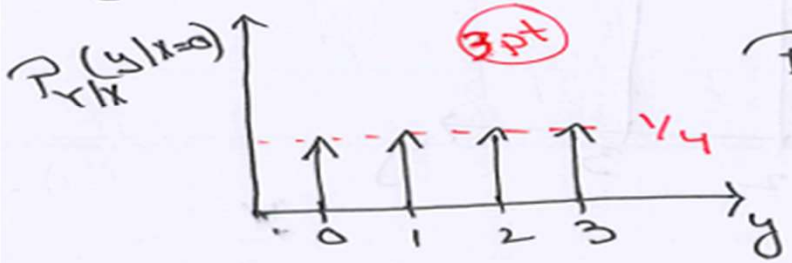
$P(X=0) = (1/2)^0 (1/2)^2 = 1/4$

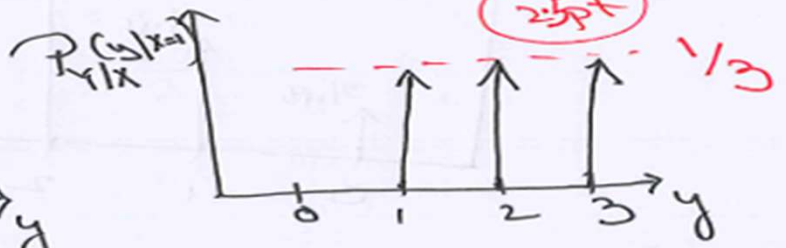
$P(X=1) = (1/2)^1 \cdot (1/2)^1 \times {}^2C_1 = 1/2$

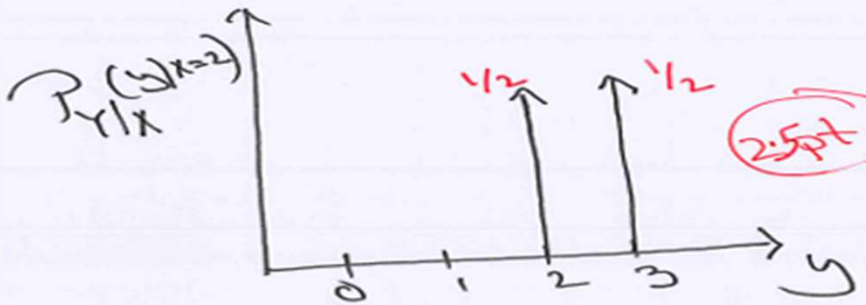
$P(X=2) = (1/2)^2 (1/2)^0 = 1/4$



(b) RV Y|X is DRV from x − 3







(c) $P_Y(y) = \sum_x P_{Y|X}(y|x) \cdot P_X(x)$ ——— 1 pt

$P_Y(0) = P_{Y|X}(0|0) \cdot P_X(0) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} = 3/48 = 0.0625$

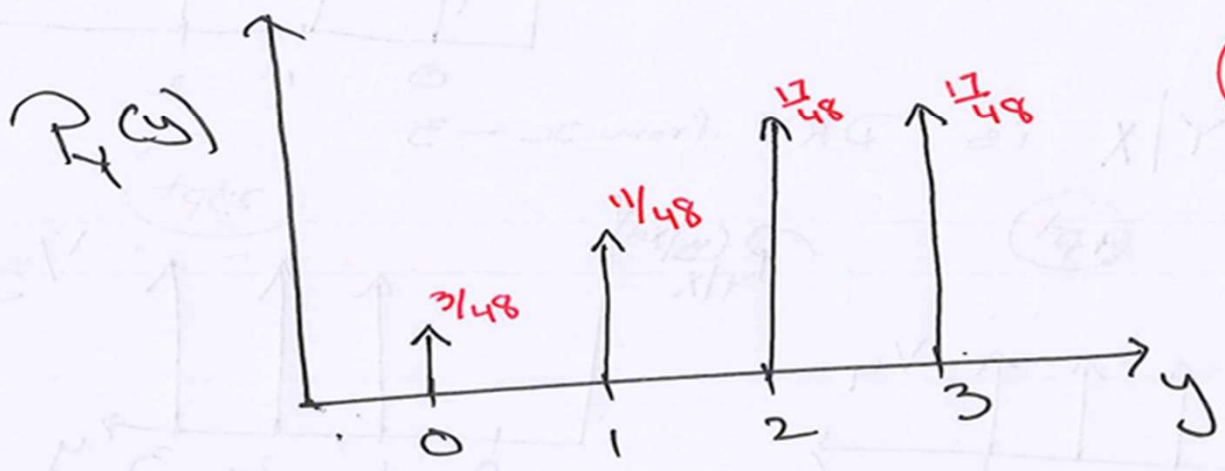
$P_Y(1) = P_{Y|X}(1|0) \cdot P_X(0) + P_{Y|X}(1|1) \cdot P_X(1) = \frac{1}{4} \cdot \frac{1}{4} + 1/3 \cdot 1/2 = \frac{11}{48} = 0.229$ 1 pt

$P_Y(2) = P_{Y|X}(2|0) \cdot P_X(0) + P_{Y|X}(2|1) \cdot P_X(1) + P_{Y|X}(2|2) \cdot P_X(2)$

$= \frac{1}{4} \cdot 1/4 + \frac{1}{3} \cdot 1/2 + 1/2 \cdot 1/4 = \frac{17}{48} = 0.354$ 1 pt

$P_Y(3) = P_{Y|X}(3|0) \cdot P_X(0) + P_{Y|X}(3|1) \cdot P_X(1) + P_{Y|X}(3|2) \cdot P_X(2)$

$= \frac{1}{4} \cdot \frac{1}{4} + 1/3 \cdot 1/2 + 1/2 \cdot 1/4 = \frac{17}{48} = 0.354$ 1 pt



4 pt

Exercise No 3

Probability and Independence

We roll a fair tetrahedral die (4-sided die with numbers 1, 2, 3 and 4) **three times**.

(a) Calculate the probability that, after three rolls, the number 4 appears at least one time. 2 Marks

In another experiment, using the same fair tetrahedral die, we roll the die **two times** and define the following two events:

- **Event A** – The sum for the two rolls is odd.
- **Event B** – The sum for the two rolls is ≥ 6.

(b) Calculate the probability $P(A \cap B)$. 2 Marks

(c) Calculate the probability $P(A \cup B)$. 3 Marks

(d) Show mathematically whether Event A and Event B are independent or not. Justify your answer. 3 Marks

(a) Probability (Number 4 appears 0 times)

$= 1 - \text{Prob}(1, 2 \text{ or } 3 \text{ appears all the 3 times})$

$= 1 - \left(\frac{3}{4}\right)^3 = 0.578$ — (2 pts)

(b) $P(A \cap B) = P(\text{Sum} = 7)$

$= P(3,4) + P(4,3)$ — (1 pt)

$= 2/16$ — (1 pt)

(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A) = P(\text{Sum is odd})$

$= P(\text{Sum}=3) + P(\text{Sum}=5) + P(\text{Sum}=7)$

$= P((1,2),(2,1)) + P((1,4),(2,3),(3,2),(4,1)) + P((3,4),(4,3))$

$= 8 \times 1/16 = \frac{8}{16} = 1/2$ — (1 pt)

$P(B) = P(\text{Sum}=6) + P(\text{Sum}=7) + P(\text{Sum}=8)$

$= P((2,4),(3,3),(4,2)) + P((3,4),(4,3)) + P((4,4))$

$= 6/16$ — (1 pt)

$P(A \cup B) = \frac{8}{16} + \frac{6}{16} - \frac{2}{16} = \frac{12}{16} = \frac{3}{4}$ — (1 pt)

(d) $P(A \cap B) = 2/16$

$P(A) \times P(B) = \frac{8}{16} \times \frac{6}{16} = \frac{3}{16}$ — (1 pt)

$\Rightarrow P(A \cap B) \neq P(A) \cdot P(B)$, thus A & B are NOT independent. — (2 pt)

Exercise No 4

You are visiting an office building for a job interview. The building has 250 workers.

To reduce the health effects of sitting all week, some employees play sports on weekends. Exactly 105 play cricket, 50 play football, 25 play table tennis, and the remaining employees play no sport. Assume these categories are mutually exclusive (no one plays more than one sport).

Among the cricket players, 30 have hypertension (high blood pressure). Among the football players, 8 have hypertension. Among the table tennis players, 12 have hypertension. Among the employees who play no sport, 60 have hypertension.

During your visit, you randomly stop a worker in the building and ask them whether they suffer from hypertension. If they answer in the positive, what is the probability they don't play any sport?

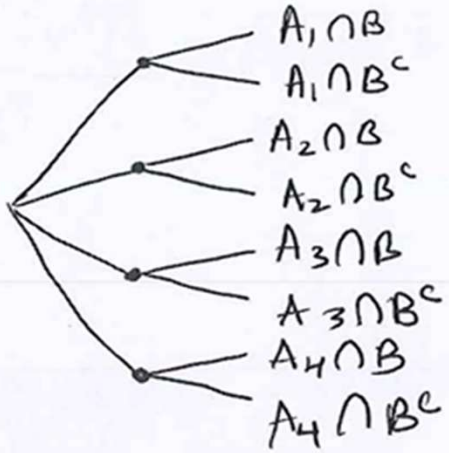
$A_1 = \{$Worker plays cricket$\}$

$A_2 = \{$ " " football $\}$

$A_3 = \{$ " " table-tennis $\}$

$A_4 = \{$ " " no sport $\}$

$B = \{$ Worker has hypertension $\}$



Using Bayes' Rule:

$$P(A_4|B) = \frac{P(A_4) \cdot P(B|A_4)}{P(B)} \quad \text{—(1)} \quad [2]$$

$$P(A_1) = \frac{105}{250}, \quad P(A_2) = \frac{50}{250}, \quad P(A_3) = \frac{25}{250}, \quad P(A_4) = \frac{70}{250}$$

$$P(B|A_1) = \frac{30}{105}, \quad P(B|A_2) = \frac{8}{50}, \quad P(B|A_3) = \frac{12}{25}, \quad P(B|A_4) = \frac{60}{70}$$

$$P(B^c|A_1) = \frac{75}{105}, \quad P(B^c|A_2) = \frac{42}{50}, \quad P(B^c|A_3) = \frac{13}{25}, \quad P(B^c|A_4) = \frac{10}{70}$$

Explicitly or Implied [4]

By Total Probability Theorem:-

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) + P(A_4 \cap B)$$
$$= P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3) + P(A_4) \cdot P(B|A_4)$$
$$= \frac{105}{250} \cdot \frac{30}{105} + \frac{50}{250} \cdot \frac{8}{50} + \frac{25}{250} \cdot \frac{12}{25} + \frac{70}{250} \cdot \frac{60}{70}$$
$$= \frac{110}{250} \quad [4]$$

From (1)

$$P(A_4|B) = \frac{P(A_4) \cdot P(B|A_4)}{P(B)} = \frac{\frac{70}{250} \cdot \frac{60}{70}}{\frac{110}{250}} = \frac{60}{110} = \frac{6}{11}$$
$$= 0.545 \quad [3]$$

For solutions skipping intermediate steps, full credit if correct, little/no partial credit if incorrect.

A Simple Alternate

$$P(A_4|B) = \frac{P(A_4 \cap B)}{P(B)}$$
$$= \frac{60/250}{\frac{110}{250}} = \frac{60}{110} = \frac{6}{11}$$
$$= 0.545$$

Exercise No 5

Consider an experiment consisting of two consecutive rolls of a fair tetrahedral die (faces labeled 1, 2, 3, and 4). The events A, B and C are defined as:

- A: The first roll is even.
- B: The second roll is even.
- C: The sum of the two rolls is even.

a) Test whether the events A, B, and C are independent. **[4 points]**

b) Test whether the events A and B are independent. **[2 points]**

c) Test whether the events A and B are independent when conditioned over C. **[4 points]**

$$\Omega = \{(1,1),(1,2),(1,3),(1,4),$$
$$(2,1),(2,2),(2,3),(2,4),$$
$$(3,1),(3,2),(3,3),(3,4),$$
$$(4,1),(4,2),(4,3),(4,4)\}$$

$$A = \{(2,1),(2,2),(2,3),(2,4),$$
$$(4,1),(4,2),(4,3),(4,4)\}$$

$$B = \{(1,2),(2,2),(3,2),(4,2),$$
$$(1,4),(2,4),(3,4),(4,4)\}$$

$$C = \{(1,1),(1,3),(2,2),(2,4)$$
$$(3,1),(3,3),(4,2),(4,4)\}$$

$P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $P(C) = \frac{1}{2}$

$P(A \cap B \cap C) = \frac{1}{4}$

$P(A \cap B) = \frac{4}{16} = \frac{1}{4}$

$P(A \cap C) = \frac{4}{16} = \frac{1}{4}$

$P(B \cap C) = \frac{4}{16} = \frac{1}{4}$

(a)

$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(B)$ ✓

$P(A \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(C)$ ✓

Pairwise independent

$P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(B) \cdot P(C)$ ✓

$P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A) \cdot P(B) \cdot P(C)$ ✗ Not Mutually Independent

[4 pts]

A, B, C are NOT Independent

(b) Since $P(A \cap B) = P(A) \cdot P(B)$, — [2 pts]

A & B are Independent

(c) $C = \{(1,1), (1,3), (2,2), (2,4)$
$(3,1), (3,3), (4,2), (4,4)\}$

$A|C = \{(2,2), (2,4), (4,2), (4,4)\}$ $\quad P(A|C) = \frac{4}{8} = \frac{1}{2}$

$B|C = \{(2,2), (2,4), (4,2), (4,4)\}$ $\quad P(B|C) = \frac{4}{8} = \frac{1}{2}$

$A \cap B|C = \{(2,2), (2,4), (4,2), (4,4)\}$ $\quad P(A \cap B|C) = \frac{4}{8} = \frac{1}{2}$

$P(A \cap B|C) = \frac{1}{2} \neq \frac{1}{2} \cdot \frac{1}{2} = P(A|C) \cdot P(B|C)$ —

A and B are NOT Independents when conditioned over C

Exercise No 6

PIN (Personal Identification Number) from a bank is a secret, numeric code that acts as your electronic signature to authenticate your identity during financial transactions, such as ATM withdrawals or point-of-sale purchases. PIN, is usually 4 to 6 digits long. The bank's current PIN selection policy is as follows:

- Length of 4 to 6 digits, i.e., user may select a 4 or a 5 or a 6-digit long PIN.
- May contain numbers (0–9)
- No spaces

a) Under the current PIN selection policy, how many possible PINs (N_1) are there? **[4 points]**

Due to recent compromises and hacking attempts, a bank is considering that PIN selection should be made more robust against hacking and decoding by intelligent code-breaking machines. The Bank's cyber security guru has proposed the following criteria for the revised PIN selection policy:

- Length of 4 to 6 characters, i.e., user may select a 4 or a 5 or a 6 characters long PIN.
- Includes at least one special character out of the 5 allowed characters, *, #, @, &, !
- No spaces
- May contain numbers (0–9), lower- and upper-case letters (a–z, A–Z)
- Is case-sensitive.

b) Under the revised PIN selection, how many possible PINs (N_2) are there? **[8 points]**

for a 4-digit Pin

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4$$
$$1 \quad 2 \quad 3 \quad 4$$

(1)

for a 5-digit Pin

$$10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 10^5$$
$$1 \quad 2 \quad 3 \quad 4 \quad 5$$

(1)

for a 6-digit pin

$$10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 10^6$$
$$1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

(

Total PINs $= 10^4 + 10^5 + 10^6$

$N_1 = 1,110,000$

(b) Possible Choices

(12pts)

$$10 + 26 + 26 + 5$$

Numbers | small alphabet | Capital alphabet | Sp Characters

$$= 67 \text{ choices} \quad \text{— (2pts)}$$

for a 4 digit PIN

$$= 67 \cdot 67 \cdot 67 \cdot 67 = 67^4 \quad \text{— (A)}$$

(A) all possible permutations

Let find without 5 special Characters

$$= 62 \cdot 62 \cdot 62 \cdot 62 = 62^4 \quad \text{— (B)}$$

(B) all permutations but No Special Characters

$$(A) - (B) = 67^4 - 62^4 \quad \text{— } (C_4) \quad \text{— (3)}$$

(C_4) → Atleast 1 Special Character

So Total PINs

$$N_2 = (67^4 - 62^4) + (67^5 - 62^5) + (67^6 - 62^6)$$